Linear Algebra I

SCIENCE PROGRAM
COURSE OUTLINE
WINTER 2026

General Information.
Discipline: Mathematics *Course code*: 201-SN4-RE
Ponderation: 2-2-2 *Credits*: 2
Prerequisite: 201-SN2-RE
Objective: To use linear algebra and vector geometry to analyze problems (0M04)

Your teacher will give you his/her schedule and availability.
Students are strongly advised to seek help promptly from their teacher if they encounter difficulties in the course.

Introduction. Linear Algebra I is the fourth Mathematics course in the Science Program. Linear Algebra, a useful tool in Physics, Chemistry and Computer Science, introduces the student to matrices and vector geometry.

The primary purpose of the course is the attainment of Objective 0M04 (to use linear algebra and vector geometry to analyze problems). To achieve this goal, the course will help the student understand the following basic concepts: systems of linear equations, matrices, determinants, two-dimensional and three-dimensional vectors from both an algebraic and a geometric perspective, applications of two-dimensional, three-dimensional, and n-dimensional vectors. The course will emphasize the ways in which geometric and algebraic concepts reinforce each other, and the ways in which these concepts can be generalized and applied to practical situations.

Emphasis is also placed on clarity and rigour in reasoning and in the application of methods. In addition to understanding and applying the basic concepts, the student will learn to prove simple propositions. This is perhaps the first course in which the student is required to produce a logical argument expressed in words, rather than just a numerical solution. The basic concepts are illustrated by applying them to various problems, which may be numerical, theoretical or applied in nature, where their application helps arrive at a solution. In this way, the course encourages the student to apply learning in one context to problems arising in another.

Textbook. The textbook for this course is *Linear Algebra with Applications (first JAC edition)*, by W. Keith Nicholson. Your teacher will provide you with the PDF, free of charge.

Course Costs. In addition to the cost of the textbook (see above), your instructor might recommend you acquire an inexpensive scientific calculator ($15–$25). *No calculators are allowed during tests or the final exam.*

Teaching Methods. This course will be 60 hours, meeting three times a week for a total of four hours a week. It relies mainly on the lecture method, although some of the following techniques are also used: question-and-answer sessions, labs, problem solving periods, class discussions, and assigned reading for independent study. No marks are deducted for absenteeism, however failure to keep pace with the lectures typically results in a cumulative inability to cope with the material and a failure in the course. A student will generally succeed or fail depending on how many problems have been attempted and solved successfully. It is entirely the student's responsibility to complete suggested homework assignments as soon as possible following the lecture, as the material will be fresher in his/her mind. This also allows the student the maximum benefit from any discussion of the homework. Answers to a selected number of problems can be found in the back of the text. Individual teachers may provide supplementary notes and problems as they see fit.

Evaluation Plan. The Final Evaluation in this course consists of the Final Exam, which covers all elements of the competency. A student's Final Grade is a combination of the Class Mark and the mark on the Final Exam. The Class Mark will be comprised of 75% tests (three or four in-class written tests) and 25% consisting of minor assessments, including at least one piece of written feedback before each test. The specifics of the Class Mark are included in an appendix that is distributed to students along with this course outline. The Final Exam is set by the SN4 course committee (which consists of all instructors currently teaching this course), and is marked by each individual instructor.

The Final Grade will be the better of:

50% Class Mark and 50% Final Exam Mark
or
25% Class Mark and 75% Final Exam Mark

A student *choosing not to write* the Final Exam will receive a failing grade of 50% or their Class Mark, whichever is less.

Students must be available until the end of the final examination period to write exams.

Other Resources.
Math Website.
http://departments.johnabbott.qc.ca/departments/mathematics

Math Study Area. Located in H-200A and H-200B; the common area is usually open from 8:30 to 17:30 on weekdays as a quiet study space. Computers and printers are available for math-related assignments. It is also possible to borrow course materials when the attendant is present.

Math Help Centre. Located near H-211; teachers are on duty from 8:30 until 15:30 to give math help on a drop-in basis.

Academic Success Centre. The Academic Success Centre, located in H-139, offers study skills workshops and individual tutoring.

College Policies.
Policy No. 7 - IPESA, Institutional Policy on the Evaluation of Student Achievement: https://www.johnabbott.qc.ca/wp-content/uploads/2021/05/Policy-No.-7-IPESA-FINAL.pdf.

Changes to Evaluation Plan in Course Outline (Article 5.3). Changes require documented unanimous consent from regularly attending students and approval by the department and the program dean.

Religious Holidays (Article 3.2.13 and 4.1.6). Students who wish to miss classes in order to observe religious holidays must inform their teacher of their intent in writing within the first two weeks of the semester.

Student Rights and Responsibilities: (Article 3.2.18). It is the responsibility of students to keep all assessed material returned to them and/or all digital work submitted to the teacher in the event of a grade review. (The deadline for a Grade Review is 4 weeks after the start of the next regular semester.)

Student Rights and Responsibilities: (Article 3.3.6). Students have the right to receive graded evaluations, for regular day division courses, within two weeks after the due date or exam/test date, except in extenuating circumstances. A maximum of three (3) weeks may apply in certain circumstances (ex. major essays) if approved by the department and stated on the course outline. For evaluations at the end of the semester/course, the results must be given to the student by the grade submission deadline (see current Academic Calendar). For intensive courses (i.e.: intersession, abridged courses) and AEC courses, timely feedback must be adjusted accordingly.

Academic Procedure: Academic Integrity, Cheating and Plagiarism (Article 9.1 and 9.2). Cheating and plagiarism are unacceptable at John Abbott College. They represent infractions against academic integrity. Students are expected to conduct themselves accordingly and must be responsible for all of their actions. For more information, see the Academic Procedure on Academic Integrity.

College definition of Cheating: Cheating means any dishonest or deceptive practice relative to examinations, tests, quizzes, lab assignments, research papers or other forms of evaluation tasks. Cheating includes, but is not restricted to, making use of or being in possession of unauthorized material or devices and/or obtaining or providing unauthorized assistance in writing examinations, papers or any other evaluation task and submitting the same work in more than one course without the teacher's permission. It is incumbent upon the department through the teacher to ensure students are forewarned about unauthorized material, devices or practices that are not permitted.

College definition of Plagiarism: Plagiarism is a form of cheating. It includes copying or paraphrasing (expressing the ideas of someone else in one's own words), of another person's work or the use of another person's work or ideas without acknowledgement of its source. Plagiarism can be from any source including books, magazines, electronic or photographic media or another student's paper or work.

Policy No. 13 - Policy on Student Conduct and Discipline Procedures: https://www.johnabbott.qc.ca/wp-content/uploads/2021/09/POLICY-13_-_policy_on_student_conduct-Final-BOG-2021-09-21.pdf.

Course Content (with selected exercises)**.** The exercises listed below should help you practice and learn the material taught in this course. Your teacher may supplement this list or assign other exercises during the semester. Regular work done as the course progresses should make it easier for you to master the course.

An item beginning with a decimal number (e. g., 3.2) refers to a section in *Linear Algebra with Applications (first JAC edition)*.

1.1 #1-14, 17
1.2 #1-5, 7-12, 15
1.3 #1-5, 6, 7, 8
1.4 #1-4
Supplementary Exercises #2, 3, 6, 11

2.1 #1-8, 10, 14-20, 24
2.2 #1-6, 8, 9, 10, 15-17, 18, 20, 21
2.3 #1-8, 16, 18, 19, 20, 21, 27, 29, 30, 31, 32, 34
2.4 #1-6, 9, 10, 11, 12-16, 18, 20, 21, 24, 25, 27, 28, 30-34, 36, 37, 38, 39, 40, 41, 42
2.5 #1, 2, 3, 5-11, 20
Supplementary Exercises #1, 2, 3, 5a, 9, 10, 13

Adjacency Matrices PDF #1-8

Torque PDF #1-5

Work PDF #1-5

3.1 #1, 2, 3, 4, 5, 6, 7, 9, 13, 14, 15, 16, 19, 22
3.2 #1, 2, 3, 5-9, 10abcdeg, 12, 13, 14, 15, 16, 17, 20, 25, 28, 29, 31, 32, 33, 34bcd
3.3 #1, 3, 8, 9, 10, 11-16, 17, 18a, 20ab, 21ab, 26, 27, 28
Supplementary Exercises #1, 4, 5

4.1 #1-4, 6-15, 17-21, 22, 23, 24, 27, 28, 29
4.2 #1, 2, 3, 4, 5, 6, 7, 9, 10-17, 18, 19, 20, 24, 30, 32, 33, 34, 35, 37, 40
4.3 # 1-5, 7, 10, 11, 16, 17

5.1 #1, 2, 3, 4, 6-10, 11, 12, 13, 14, 15-22
5.2 #1-6, 7, 8, 9, 10-15, 17-20
5.3 #6, 8, 12, 13, 14
5.4 #1, 2, 3, 5, 6, 7, 9, 10, 11, 14, 15
Supplementary Exercises #1

OBJECTIVE	STANDARD
Statement of the Competency To use linear algebra and vector geometry to analyze problems (0M04).	**Performance Criteria for the Competency as a Whole** • Correct use of mathematical terminology and syntax. • Algebraic manipulations conforming to established rules. • Appropriate use of required technological tools. • Demonstration of rigorous mathematical reasoning through the use of concepts, properties and theorems.
Elements of the Competency	**Performance Criteria**
1. Use matrix language.	• Correct recognition of particular matrices. • Appropriate operations on matrices. • Appropriate use of the properties of determinants. • Exact calculation of the determinant of a matrix.
2. Manipulate geometric and algebraic vectors in the plane and space.	• Correct distinction of the characteristics of vectors. • Appropriate operations on vectors. • Appropriate graphical representation of vectors in the plane and in space. • Correct interpretation of linear independence and dependence of vectors. • Appropriate use of a basis. • Correct determination of the orthogonal projection. • Correct determination of products of vectors.
3. Represent lines and planes in Euclidean space.	• Correct determination of equations of lines and planes. • Appropriate graphical representation of lines and planes. • Accurate determination of the relative position between straight lines and planes.
4. Use methods from linear algebra and vector geometry in mathematical applications.	• Correct determination of the inverse of a matrix. • Correct use of matrix methods for solving systems of linear equations. • Correct interpretation of types of solutions of systems of linear equations. • Exact calculation of distances and measures of angles. • Correct determination of the intersection between lines and planes. • Rigorous demonstrations of propositions.
5. Apply the concepts of linear algebra and vector geometry to solving problems in the natural sciences.	• Rigorous use of linear algebra and vector geometry methods. • Correct resolution of problems by the use of matrices. • Correct resolution of problems by the use of vectors. • Relevant application of the products of vectors. • Correct interpretation of the results.

Specific Performance Criteria	Intermediate Learning Objectives
1. *Matrices*	
1.1 Correct recognition of particular matrices.	1.1.1. Identity, zero, symmetric, skew-symmetric, diagonal, triangular, row-echelon, reduced row-echelon, elementary, augmented, coefficient, commutative.
1.2 Appropriate operations on matrices.	1.2.1. Addition, scalar multiplication, matrix multiplication, inversion, transpose. 1.2.2. Diagonalization of 2×2 and 3×3 matrices over the real numbers.
1.3 Exact calculation of the determinant of a matrix.	1.3.1. Determinant of a 2×2 matrix. 1.3.2. Determinant of a matrix by cofactor expansion.
1.4 Appropriate use of the properties of determinants.	1.4.1. Simplifying determinants via row and column operations. 1.4.2. The adjugate of a matrix. 1.4.3. The inverse of a matrix in terms of its determinant and adjugate. 1.4.4. Cramer's Rule.
2. *Vectors*	
2.1 Correct distinction of the characteristics of vectors.	2.1.1. Length and direction. Parallel and orthogonal vectors. Unit vectors.
2.2 Appropriate operations on vectors.	2.2.1. Addition, subtraction, scalar multiplication, matrix multiplication, linear combination, normalization. 2.2.2. The span of a set of vectors in $\mathbb{R}^n$.
2.3 Appropriate graphical representation of vectors in the plane and in space.	
2.4 Correct interpretation of linear independence and dependence of vectors.	
2.5 Appropriate use of a basis.	2.5.1. The dimension of a subspace of $\mathbb{R}^n$.
2.6 Correct determination of the orthogonal projection.	
2.7 Correct determination of products of vectors.	2.7.1. Dot product, scalar product, triple scalar product. 2.7.2. Areas of parallelograms and triangles in $\mathbb{R}^3$. 2.7.3. Volume of a parallelepiped.
3. *Lines and planes*	
3.1 Correct determination of equations of lines and planes.	3.1.1. Equations of a line: vector, parametric, and symmetric. 3.1.2. Equations of a plane: vector, parametric, and normal.
3.2 Appropriate graphical representation of lines and planes.	
3.3 Accurate determination of the relative position between straight lines and planes.	
4. *Applications (mathematical)*	
4.1 Correct determination of the inverse of a matrix.	
4.2 Correct use of matrix methods for solving systems of linear equations.	4.2.1. Elementary row operations. 4.2.2. Row reduction, inverse matrix. 4.2.3. Homogeneous systems, trivial and non-trivial solutions.
4.3 Correct interpretation of types of solutions of systems of linear equations.	4.3.1. No solution, unique solution, or infinitely many solutions. 4.3.2. Solution sets: empty, singleton, infinite. 4.3.3. Rank, dimension of null space, and the relation between them. 4.3.4. A linear system and the corresponding vector and matrix equations. 4.3.5. Column space and null space of a matrix.
4.4 Exact calculation of distances and measures of angles.	4.4.1. Distances: point and a point, point and a line, point and a plane. 4.4.2. The angle between two vectors.
4.5 Correct determination of the intersection between lines and planes.	
4.6 Rigorous demonstrations of propositions.	
5. *Applications (natural sciences)*	
5.1 Rigorous use of linear algebra and vector geometry methods.	
5.2 Correct resolution of problems by the use of matrices.	5.2.1. Balance chemical equations. 5.2.2. Adjacency matrices. Walks of length n in a graph.
5.3 Correct resolution of problems by the use of vectors.	
5.4 Relevant application of the products of vectors.	5.4.1. Work and torque.
5.5 Correct interpretation of the results.	